Group Theory Notes

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Outline

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Introduction

Examples of Groups

Permutations

Rubik's cube group

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Example (Rubik's cube)

Consider the Rubik's cube (for an online example, see here). What moves can we perform on a Rubik's cube?

Rules of moves

There are many, *many* different configurations (or states) possible, but only one solution. We can get from one state to another by performing a valid movement of the cube. These movements have the following properties:

1. There is an unchanging list of allowable moves.
2. Every move is reversible.
3. Every move is deterministic.
4. Moves can be combined.

Rubik's Cube

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Rubik's Cube

Example (Generators of Rubik's cube motions)

The website above uses twelve basic moves: F, F′, U, U′, and so on.

- How can you write F′ in terms of F?
- What is the smallest number of moves you can find that represent all of the basic moves on a Rubik's cube?

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Groups

The observations above form the basis of a mathematical structure called a group. Groups appear throughout mathematics and its applications.

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Examples of Groups

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Swaps

Example (Coins in a line)

Consider placing a penny and a nickel side by side in front of you. Is the action of swapping the positions of both coins a group? Note that you will need to specify an appropriate list of moves and determine if that list satisfies the above rules.

Example (Marbles in piles)

Suppose you have five marbles in each of two piles in front of you. You have two actions available to you: you can take a marble from the left pile and move it to the right pile, *or* you can take a marble from the right pile and add it to the left pile. Is this a group?

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Actions on Words

Example (Three letter words)

Consider all possible three letter "words" that can be formed from the letters $\{A, B, C\}$, where each letter must be used precisely once.

1. How many words are possible?
2. Consider two actions: swapping the first and second letter of a word, and swapping the second and third letter of a word. Can these two actions alone generate all of the words possible starting from ABC?
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End of class *<2026-06-22 Mon>*

Group Theory

At this point, you might start thinking of a group as a set of objects that we can move around in a nice way. Group theory is about the study of such sets following the rules outlined above.

Example (Coins in a line (again))

Consider the group of coins in a line.

1. By rule 4, the action from this exercise when performed twice in a row yields another action in the group. What is this action?
2. Does every group have an action like this?

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1. By rule 4, the action from this exercise when performed twice in a row yields another action in the group. What is this action?
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As a reminder, the group rules are as follows:

1. There is an unchanging list of moves.
2. Every move is reversible.
3. Every move is deterministic.
4. Moves can be combined.

Example (Symmetry groups)

Describe the symmetry group of an equilateral triangle and a non-square rectangle.

Examples of Groups

Example (Symmetry groups)

Describe the symmetry group of an equilateral triangle and a non-square rectangle.

Recall that the symmetry group of a shape is the set of reflections/rotations that leaves the shape unchanged.

Groups with Certain Properties

Example (Groups with restrictions)

Try to find or construct groups that satisfy the following constraints:

1. The order in which the actions are performed matters.
2. The order in which the actions are performed doesn't matter.
3. There are exactly three actions.
4. There are exactly four actions.
5. There are infinitely many actions.

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1. Rubik's cube group or symmetry group of triangle/square
2. Symmetry group of non-square rectangle or coin group
3. Look at triangle
4. Look at square or non-square rectangle
5. Can use the integers or an infinite chain

Nonexamples of Groups

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Devise a situation that satisfies all of the group rules except for the second, third or fourth depending on your group.

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Break class into three groups (count off 1, 2, 3) and have the groups focus on breaking a single rule from group rules.

1. Breaks second rule: can use naturals with addition or an action that "erases" information.
2. Think of symmetry group where we rotate or flip based on results of coin toss.
3. We already have an example of this. We can also look at limited symmetry groups (i.e., subgroups that aren't closed).

Permutations

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Permutations

- A permutation of an ordered collection of objects is a rearrangement of those objects.
- Every group action we've looked at so far is an example of a permutation.

Example (Permutations and groups)

List the permutations and the objects being permuted for the following groups:

1. The three letter word group (see Example (Three letter words)).
2. The symmetry group of the equilateral triangle (see Symmetry Groups (slide)).

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Symmetry Groups

- Shapes have symmetry groups (also called *dihedral groups*).
- Collections of objects also have symmetry groups. The actions are permutations.

Example (Elements of a symmetry group)

Give two elements of the symmetry group of $\{1, 2, 3, 4\}$.

Example (Number of elements of a symmetry group)

How many elements are in the symmetry group above?

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Example (Number of elements of a symmetry group)
How many elements are in the symmetry group above?

Note the connection to previous permutations!

- Two permutations on the same set of elements can be combined.

Example (Combining permutations)

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 4 & 6 & 2 & 1 \end{pmatrix}, \rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 6 & 2 & 5 \end{pmatrix}$$

Find $\sigma\rho$.

Composition of Permutations

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Cycle Notation

- The permutation $\sigma \in S_3$ given by

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can be written as (231).

More cycles

Write the permutations σ and ρ from Example: Combining permutations using cycle notation.

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S_n

- The symmetry group on n elements is called S_n.
- You've already found S_3!
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Cycle Notation Again

Example (Cycle notation in S_4)

Write the permutations you found in Example (Elements of a symmetry group) using cycle notation.

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Combining Permutations Again

- Cycle notation provides a convenient way to combine permutations.

Example (Combining permutations with cycles)

Find $\sigma\rho$ using cycle notation.

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Cayley Tables

A Cayley table is a multiplication table for a group.

Example (Cayley table for symmetry group)

Find the Cayley table for the symmetry group of the rectangle

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Even and Odd Permutations

- Permutations can be categorized as even or odd based on how many switches they perform
- This is called the parity of the permutation

Example (Determine parity in S_3)

Find the parity of the permutations (12) and (123).

Example (Determine parity in S_6)

Find the parity of the permutations σ, ρ and $\sigma\rho$.

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Permutation Puzzles

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- Knowing something about the allowed permutations tells us about the puzzle itself!
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Mutilated Chessboard Puzzle

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with two opposite corners removed using 2×1 dominoes.

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Mutilated Chessboard Puzzle

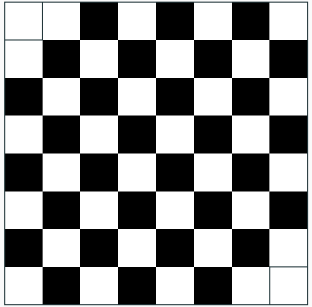

Figure 1: Chessboard with two corner pieces missing.

Mutilated Chessboard Puzzle

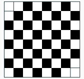

Figure 1: Chessboard with two corner pieces missing.

Invariants and Solvability

- No matter how you place dominoes on the chessboard, there is something that must be true of each domino. This is an invariant of the puzzle.

Example (Impossibility of solution)

Explain why the chessboard puzzle is not solvable. *Hint*: find an invariant for the puzzle, i.e., something that must be true no matter how you place dominoes on the puzzle.

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The 15-Puzzle

- The goal is to solve the puzzle to put the numbers in order (i.e., swap the 14 and 15 squares)
- The blank square must be on the bottom right at the end

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

Figure 2: The 15-puzzle

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13	15	14	

Figure 2: The 15-puzzle

"A Pestilence"

FIFTEEN.

No pestilence has ever visited this or any other country which has spread with the awful celerity of what is popularly called the "Fifteen Puzzle." It is only a few months ago that it made its appearance in Boston, and it has now spread over the entire country. Nothing arrests it. Neither age nor sex is spared by it, and it now threatens our free institutions, inasmuch as from every town and hamlet there is coming up a cry for a "strong man" who will stamp out this terrible puzzle at any cost of Constitution or freedom.

In the presence of this giant evil, all our customary defenses prove valueless. The Police cannot arrest a seller or a victim of the puzzle, since the law knows nothing of it. Mr. COMSTOCK has in vain tried to find something in it which would warrant him in attempting to suppress it. The pulpit and the press set forth its dangerous nature, but no one heeds them, and even the various Societies for the Prevention of Different Things seem utterly powerless.

Figure 3: Excerpt from New York Times, 1880

"A Pestilence"

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Example (Model a configuration)

Use a permutation to describe the following arrangement of the 15-puzzle:

8	12	2	5
11	1	6	
7	14	10	15
9	4	3	13

Figure 4: A shuffled 15-puzzle

Permutations and the 15-Puzzle

Example (Model a configuration)

Use a permutation to describe the following arrangement of the 15-puzzle:

8 12 2 5
11 1 6
7 14 10 15
9 4 3 13

Figure 4: A shuffled 15-puzzle

The allowed moves of the 15-puzzle do *not* form a group!

Parity Again

Example (Find parity of configurations)

Determine the parities of the permutations corresponding to the puzzles in the previous examples, and find the parity of the solution of the 15-puzzle.

Parity of moves in the 15-puzzle

Every valid configuration in the 15-puzzle (that is, moves that place the empty spot in position 16) corresponds to an even permutation.

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Impossibility of the 15-Puzzle

Theorem (Nonexistence of solution)

There is no solution of the 15-puzzle.

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Example (Empty square on bottom left)

Suppose we don't want the empty square to end up on the bottom right, but rather the bottom left:

1	2	3	4
5	6	7	8
9	10	11	12
	13	14	15

Figure 5: The modified 15-puzzle.

Is the puzzle now solvable?

Variation of the 15-Puzzle
Example (Empty square on bottom left)
Suppose we don't want the empty square to end up on the bottom right, but rather the bottom left:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15
Figure 5: The modified 15-puzzle.
Is the puzzle now solvable?

Rubik's cube group

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Terminology

The cubies on a Rubik's cube can be placed in the following categories:

1. Corner cubies
2. Edge cubies
3. Center cubies

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Cubies and Cubicles

- A cubicle is the position of a given cubie. Moves changes the positions of the cubies, but *not* the cubicles.
- Positions can be specified by listing incident faces: ufr represents the cubicle/cubie on the upper front right face (i.e., a corner).

Example (Edge Positions)

Give all positions for edge cubies on the back face of a cube.

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Example (Generators Acting on Cubies)

What happens to cubies in the previous categories if you apply any of the six basic Rubik's cube actions?

Group Actions and Cubies
Example (Generators Acting on Cubies)
What happens to cubies in the previous categories if you apply any of the six basic Rubik's cube actions?

Estimating Rubik's Cube Group

Example (Size of the Rubik's cube group)

The Rubik's cube group is based on the six fundamental motions F, B, U, D, L and R. How large could this group be? *Hint*: try answering the following questions.

1. How many different positions can the corner cubies be in?
2. How many different positions can the edge cubies be in?

Estimating Rubik's Cube Group

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1. How many different positions can the corner cubies be in?
2. How many different positions can the edge cubies be in?

1. The corner cubies can be arranged in up to $8!$ ways with up to 3 different configurations each, giving an upper bound of $8!3^8$ arrangements for the corner cubies.
2. The edge cubies have an upper bound of $12!2^{12}$ arrangements.
3. The upper bound for moves is therefore $8!3^8 12!2^{12}$.

Implementing the Cube Group

- The cube group is generated by the actions F, B, U, D, L, R.

Example (Describing actions)

Give the permutations that correspond to the above actions.

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Work through permutation for B as a class as follows:

1. List all numbers on layer back layer.
2. Go through and trace the path of one number at a time. Remove numbers from list as you do.
3. The answer is

$$B = (1, 14, 48, 27)(2, 12, 47, 29)(3, 9, 46, 32)(33, 35, 40, 38)(34, 37, 39, 36)$$